IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

# Charging Infrastructure

## Lecture-55

## DC Charging System(BEVC-DC001)

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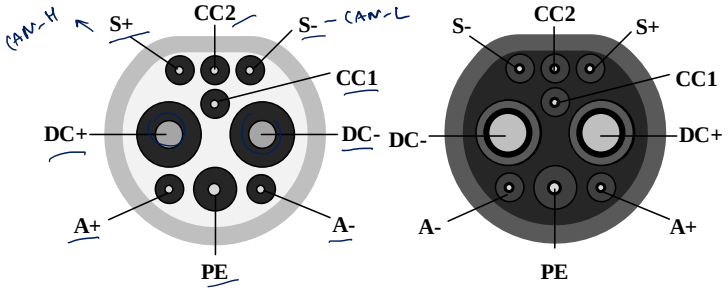


# Recap

- CCS2 DC charging system: Charging states, PP description, CP description, Charging sequence, PLC, PLC initialization, HLC with CCS2

# BEVC-DC001

- Input: 3-phase AC
- Output: Supports up to a maximum of 10kW for 48V system and 15 kW for 72V systems
- Communication is derived from IEC 61851-24B and GB/T 27930
- Uses CAN communication
- GB/T 20234.3 Connector with 5 m Cable




Charger Plug

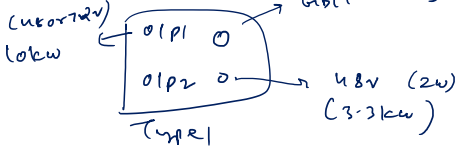


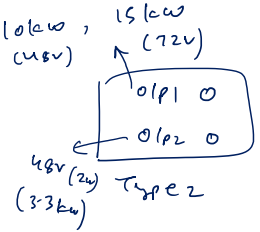

Vehicle inlet connector

**Introduced by Department of Heavy Industries, Government of India**

https://heavyindustries.gov.in/sites/default/files/2023-09/Standardization%20of%20protocol.pdf

# BEVC-DC001 Cont.




- EVSE has 2 no. of outputs
- Mode-4
- v/g measurement = ±0.5%
- Current measurement = ±1A, ≤50A
- Protection against overvoltage <1A within 2s, if v/g exceeds the max. limit of battery voltage
- EV - EVSE → (CAN communication)
- EVSE - CMS → OCPP (open charge Point Protocol)
- PCU → η > 92% (at nominal Power output)
  PF ≥ 0.9 (full load)
  Max current = 200A
- Tolerance for the v/g & current regulation at the output → { ±2.5A, <50A } CC charging
  { ±5%, >50A }
  { Max. 2% for the max. rated voltage of EVSE } CV charging

https://heavyindustries.gov.in/sites/default/files/2023-09/Standardization%20of%20protocol.pdf

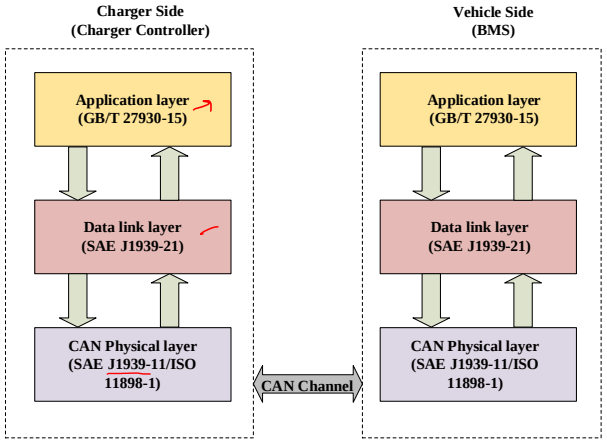
# BEVC-DC001 Cont.

- The digital communication happens over the Controlled Area Network (CAN).
- For the CAN bus line (CAN_H and CAN_L) uses shielded twisted pair wires with a termination resistance of 120Ω.
- Proximity detection is done via charging confirmation pins which provides the simple hardware based confirmation indicating that the physical connection is being done. It also ensures the locking of the charger plug into the vehicle.

https://heavyindustries.gov.in/sites/default/files/2023-09/Standardization%20of%20protocol.pdf

# Communication in BEVC-DC001 (Derived from GB/T 27930)

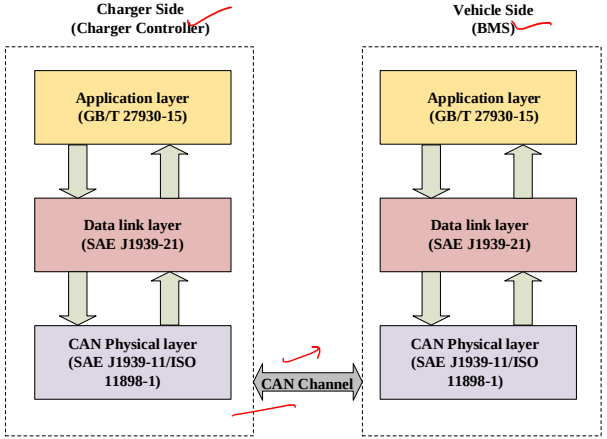
- It defines rules for implementing the physical and data link layers based on SAE J1939 standard.
- Physical layer defines: termination conditions, voltage levels, differential signaling, speed, etc.
- Data link layer defines the addressing for the communication nodes and CAN data frame formats.
- The application layer of GB/T 27930-2015 defines representation of parameters and their parameter group numbers.
- CAN data frames are transmitted in the form of Protocol Data Unit (PDU) are also defined in GB/T 27930.
- Speed – 250Kbps and uses CAN 2.0B




Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

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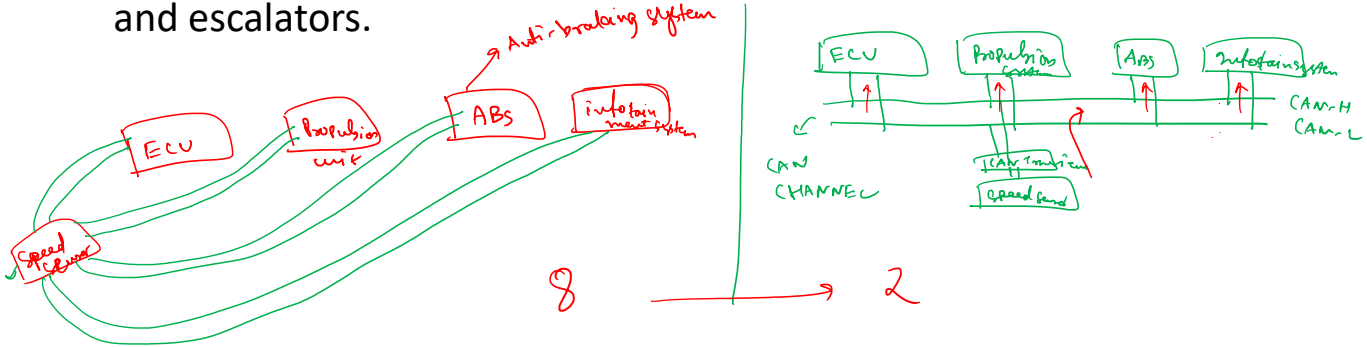
- Only two participants at a time – a charger and a BMS in the vehicle – with fixed addresses:
  - 86 (56Hex) for the Charger
  - 244 (F4Hex) for the BMS
- No direct connections to other CAN systems in the vehicle, such as the power train CAN.




Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

# CAN Communication

- The Controlled Area Network (CAN) is a communication network between the devices.
- CAN communication is done with network comprises of nodes and is a message-based protocol.
- It is a two-wire serial bus communication and messages are sent sequentially.
- CAN was developed to reduce cable wiring, so the separate control units inside a vehicle could communicate with only a single pair of wires.
- Message-based protocol also allows for several features to be added via software alone.
- CAN based communication is widely used in automotive industries, avionics systems, medical systems and escalators.

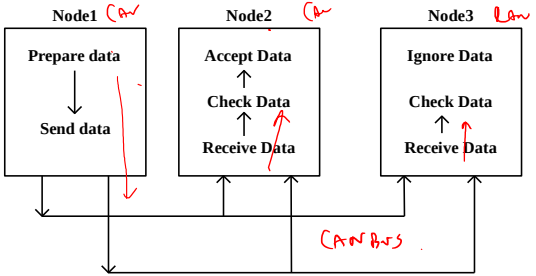


# CAN Communication Benefits

- Low cost
  - Controller units communicate with a single CAN interface reducing cables.
- Centralized communication
  - Central diagnosis and configurations among all the controller unit
- Robust
  - CAN signals are robust against electromagnetic interferences (differential signaling).
- Efficient
  - CAN messages are prioritized by IDs, and top priority gets the access of the CAN bus.
- Error Capability
  - CAN data frame has cyclic redundancy check (CRC) code to perform error checking on each frame's content.
  - An error frame is transmitted to indicate the occurrence of the error. Frames with error can be discarded by all nodes.

# CAN Communication Benefits Cont.

- Flexible
  - Each CAN-connected controller receives all transmitted messages on the CANBUS and decides its relevance (by comparing it with the CAN database) and acts accordingly. This allows easy modification and inclusion of additional nodes
- Speed
  - Supports data rate of 1Mbps till 40m, 125Kbps till 100m distance
- Multi-master configuration
  - Any node can access the bus if it is free.



# Thank You